Omkar Thawakar

Abu Dhabi, UAE | +971-585-25-1761 | thawakar.omkar@gmail.com |
omkarthawakar.github.io | LinkedIn | GitHub | Google Scholar

Research Interests

Multimodal AI • Large Multimodal Models (LMMs) • Vision-Language Models (VLMs) • Computer Vision • Video Understanding • Visual Reasoning • Multimodal Reasoning • AI Agents • Self-Evolving AI Systems • Generative UI Systems • Video Retrieval • Image & Video Segmentation

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Abu Dhabi, UAE
Ph.D. in Computer Vision; GPA: 3.74 / 4.00 — *Aug. 2023 – Spring 2027*

- Thesis: Video Understanding with Large Multimodal Models. Advisor: Prof. Fahad Khan.
- Research: Multimodal Learning, Vision-Language Models, LLMs, LMMs, Agents, Video Understanding.

Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T) — Nanded, India
B.Tech. in Computer Science Engineering; GPA: 8.7 / 10.0 — *Jun. 2015 – Jul. 2019*

- Coursework: Neural Networks, Machine Learning, Algorithms, Operating Systems, Cryptography, Distributed Systems, Computer Networking.
- Awarded Best B.Tech Thesis Award.

Research & Industry Experience

Adobe Research — 2026 – Ongoing
Research Scientist Intern — *India*

- Developing self-evolving multimodal agents that generate, evaluate, and iteratively improve interactive GUI experiences using feedback from earlier generations.
- Designing a benchmark and evaluation suite for GUI generation covering functionality, prompt alignment, visual quality, content grounding, component coverage, and interaction.
- Building evolutionary-memory that convert failed generations into reusable rules and improvement signals.

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — May 2021 – Jul. 2023
Researcher — *Abu Dhabi, UAE*

- Developed a multi-scale spatio-temporal transformer for video instance segmentation, achieving state-of-the-art performance and publication at ECCV 2022.
- Designed a transformer-based method for 3D mitochondria instance segmentation, improving accuracy by approximately 3% over prior approaches.
- Introduced an open-world video segmentation framework for detecting and segmenting previously unseen object categories, later published in IJCV.

Chefling India Pvt. Ltd. — Feb. 2020 – May 2021
Machine Learning Engineer — *Bangalore, India*

- Deployed an ML-assisted recipe extraction pipeline using Scrapy, achieving 87% accuracy across websites with diverse and irregular structures.
- Built a recipe classification service covering 70+ categories, replacing an external provider and reducing operating costs by 92%.

Indian Institute of Technology Ropar (IIT-Ropar) — Jan. 2019 – Feb. 2020
Research Assistant — *Ropar, India*

- Researched video super-resolution, moving-object segmentation, dehazing, low-light enhancement, and image restoration under adverse conditions.
- Developed a recurrent GAN for video super-resolution.

Applied AI Leadership

AI Technical Lead — Jul. 2024 – May 2026
Lawa.AI — *Abu Dhabi, UAE*

- Led the design and deployment of production-grade agentic AI systems, including Lawa-Agent-Studio and RAG-based enterprise AI applications.
- Developed AI products from research prototypes to scalable agentic AI pipelines and represented UAE and MBZUAI AI innovation at ITU WTSA 2024 Innovation Xchange.

AI Systems Lead — Jan. 2024 – Dec. 2025
Nutrigenics.Care — *Abu Dhabi, UAE*

- Developed Nutrigenics, an AI platform for clinical nutrition intelligence and personalized healthcare.
- Secured competitive innovation funding, including Microsoft Founders Hub support, culminating in technology acquisition by a healthcare entity.

Publications

(* indicates joint first authors, + indicates my role as mentor)

- [**Under Review**] **Omkar Thawakar**, Shravan Venkatraman, Ritesh Thawkar, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, and Fahad Khan. *"Self-Evolving Game World."*
- [**Under Review**] Ritesh Thawkar, Shravan Venkatraman, **Omkar Thawakar+**, Abdelrahman Shaker, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, and Fahad Khan. *"Ask, Solve, Generate: Self-Evolving Unified Multimodal Understanding and Generation via Self-Consistency Rewards."*
- [**ECCV 2026**] Shravan Venkatraman, Ritesh Thawkar, **Omkar Thawakar+**, Rao Muhammad Anwer, Hisham Cholakkal, Salman Khan, Fahad Khan *"Paying More Attention to Visual Tokens in Self-Evolving Large Multimodal Models"*
- [**Under Review**] **Omkar Thawakar**, Shravan Venkatraman, Ritesh Thawkar, Abdelrahman Shaker, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, and Fahad Khan. *"EvoLMM: Self-Evolving Large Multimodal Models with Continuous Rewards."*
- [**Under Review**] **Omkar Thawakar**, Dmitry Demidov, Vaishnav Potlapalli, Sai Prasanna Teja Reddy Bogireddy, Viswanatha Reddy Gajjala, Alaa Mostafa Lasheen, Rao Muhammad Anwer, and Fahad Khan. *"CoVR-R: Reason-Aware Composed Video Retrieval."*
- [**CVPR 2026**] Dmitry Demidov, Zaigham Zaheer, Zongyan Han, **Omkar Thawakar**, and Rao Muhammad Anwer. *"Thinking Beyond Labels: Vocabulary-Free Fine-Grained Recognition using Reasoning-Augmented LMMs."*
- [**ICCV 2025**] **Omkar Thawakar**, Dmitry Demidov, Ritesh Thawkar, Rao Muhammad Anwer, Mubarak Shah, Fahad Shahbaz Khan, and Salman Khan. *"Beyond Simple Edits: Composed Video Retrieval with Dense Modifications."*
- [**ACL 2025**] **Omkar Thawakar**, Dinura Dissanayake, Ketan More, Ritesh Thawkar, Ahmed Heakl, Noor Ahsan, Yuhao Li, Mohammed Zumri, Jean Lahoud, Rao Muhammad Anwer, Hisham Cholakkal, Ivan Laptev, Mubarak Shah, Fahad Shahbaz Khan, and Salman Khan. *"MobiLlama: Towards Accurate and Lightweight Fully Transparent GPT."*
- [**ICLR 2025 Spotlight**] **Omkar Thawakar**, Ashmal Vayani, Salman Khan, Hisham Cholakkal, Rao M. Anwer, Michael Felsberg, Tim Baldwin, Eric P. Xing, and Fahad Shahbaz Khan. *"MobiLlama: Towards Accurate and Lightweight Fully Transparent GPT."*
- [**CVPR 2025 Highlight**] Ashmal Vayani, Dinura Dissanayake, Hasindri Watawana, Noor Ahsan, Nevasini Sasikumar, **Omkar Thawakar+**, et al. *"All Languages Matter: Evaluating LMMs on Culturally Diverse 100 Languages."*
- [**NAACL 2025**] Sara Ghaboura, Ahmed Heakl, **Omkar Thawakar+**, Ali Alharthi, Ines Riahi, Abduljalil Saif, Jorma Laaksonen, Fahad S. Khan, Salman Khan, and Rao M. Anwer. *"CAMEL-Bench: A Comprehensive Arabic LMM Benchmark."*
- [**CVPR 2024**] **Omkar Thawakar**, Muzammal Naseer, Rao Muhammad Anwer, Salman Khan, Michael Felsberg, Mubarak Shah, and Fahad Shahbaz Khan. *"Composed Video Retrieval via Enriched Context and Discriminative Embeddings."*
- [**ACL-W 2024**] **Omkar Thawakar**, Abdelrahman Shaker, Sahal Shaji Mullappilly, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Jorma Laaksonen, and Fahad Shahbaz Khan. *"XrayGPT: Chest Radiographs Summarization using Large Medical Vision-Language Models."*
- [**IJCV 2024**] **Omkar Thawakar**, Sanath Narayan, Hisham Cholakkal, Rao Muhammad Anwer, Salman Khan, Jorma Laaksonen, Mubarak Shah, and Fahad Shahbaz Khan. *"Video Instance Segmentation in an Open-World."*
- [**MICCAI 2023**] **Omkar Thawakar**, Rao Muhammad Anwer, Jorma Laaksonen, Orly Reiner, Mubarak Shah, and Fahad Shahbaz Khan. *"3D Mitochondria Instance Segmentation with Spatio-Temporal Transformers."*
- [**ECCV 2022**] **Omkar Thawakar**, Sanath Narayan, Jiale Cao, Hisham Cholakkal, Rao Muhammad Anwer, Muhammad Haris Khan, Salman Khan, Michael Felsberg, and Fahad Shahbaz Khan. *"Video Instance Segmentation via Multi-Scale Spatio-Temporal Split Attention Transformer."*

Patents

- [**US Patent**] **Omkar Thawakar**, Sanath Narayan, Jiale Cao, Hisham Cholakkal, Rao Muhammad Anwer, Muhammad Haris Khan, Salman Khan, Michael Felsberg, and Fahad Shahbaz Khan. *"System and Method for Video Instance Segmentation via Multi-Scale Spatio-Temporal Split Attention Transformer."* Patent #12499573, Published Dec. 2025.
- [**US Patent**] **Omkar Thawakar**, Alexandre RIVKIND, Ehud Ahissar, Fahad Khan. *"System and method for video instance segmentation via recurrent encoder-based transformers"* Patent #18435537, Published August. 2025.

Technical Skills

Research Areas: Computer Vision, Multimodal Learning, Vision-Language Models, Video Understanding, LLMs, Agentic AI, Dense Retrieval, Image/Video Segmentation
Deep Learning: PyTorch, HuggingFace Transformers, LoRA/QLoRA, RLHF, DPO, RAG, Diffusion Models
Programming: Python, R, Java, C
MLOps / Cloud: AWS, Azure, Google Cloud Platform, Docker, MLflow, BigQuery
Data & Web: PySpark, NumPy, Pandas, Matplotlib, Scikit-learn, Flask, Django, React-JS, MongoDB, SQLite

Honors & Awards

- Winner: National Grant Program for Culture & Creativity, 100K AED, Ministry of Culture, Nov. 2025.
- 1st Place: Khalifa Fund Innovation Competition, 250K AED product grant, ADBW, Dec. 2024.
- ICLR-W 2025 Spotlight: MobiLlama selected as Top 2% paper at SLLM Workshop, Feb. 2025.
- Microsoft Founders Hub Grant: Nutrigenics.Care awarded $150K in technical support, Jul. 2024.
- 1st Place: Sandook-Al-Watan Student Pitch Competition, 30K AED project grant, Nov. 2024.
- 1st Place: Pitch Day with Microsoft, MBZUAI Incubation & Entrepreneurship Center, May 2024.
- Best B.Tech Thesis Award: Video Super-Resolution using Recurrent GAN, SGGSIE&T, May 2019.
- Best Research Paper: IC3NS-2018, Application of ML Algorithms for IPM Device, May 2018.

Teaching and Academic Services

Teaching | MBZUAI

- Teaching Assistant for graduate-level Computer Vision courses at MBZUAI during Aug. 2023 – Dec. 2025, including support for lectures, hands-on exercise and tutorials, assignments, lab sessions and student projects.
- Provided technical guidance on computer vision, multimodal learning, deep learning, and vision-language models.

Conference Services

- **Organized Composed Video Retrieval Challenge** at CVPR with Video Large Language Model Workshop during the year 2025-2026.
- **Reviewer:** CVPR, ICCV, ECCV, IEEE, CVF, and Springer venues; reviewed 200+ research papers since Jul. 2022.
- Reviewed submissions belongs to multimodal learning, video understanding, image/video segmentation, large language models, large multimodal models, and reinforcement learning, etc.

Research Mentorship

- Mentored junior researchers and students on multimodal reasoning, video understanding, retrieval, and LLM projects.
- Provided guidance on research problem formulation, dataset construction, model development, experimentation, and academic research paper/thesis writing.
- Supported collaborative research projects leading to submissions and publications at top-tier CV and NLP venues.

Invited Talks and Research Presentations

- Presented CoVR-R at 2nd Workshop on Video LLM at CVPR-2026.
- Represented MBZUAI and UAE AI innovation at the ITU WTSA 2024 Innovation Xchange.

References

Prof. Fahad Khan: Primary PhD Advisor, Deputy Chair CV, MBZUAI, fahad.khan@mbzuai.ac.ae
Prof. Timothy Baldwin: Mentor / Collaborator, Provost, MBZUAI, timothy.baldwin@mbzuai.ac.ae
Prof. Mubarak Shah: Mentor / Collaborator, University of Central Florida, shah@crcv.ucf.edu
Balaji Vasan Srinivasan: Line Manager during Internship, Principle Scientist, Adobe, balsrini@adobe.com